

DEPARTAMENTO DE DESARROLLO HUMANO

UNIVERSIDAD YACHAY TECH

Scientific dissemination

Andrés Tirado, Ph.D.

www.yachaytech.edu.ec

Scientific communication channels

Formal channels

They are communications of a public and lasting nature.

The material is available to a wide audience.

Scientific journals, books, reports and proceedings.

Informal channels

Ephemeral forms of communication.

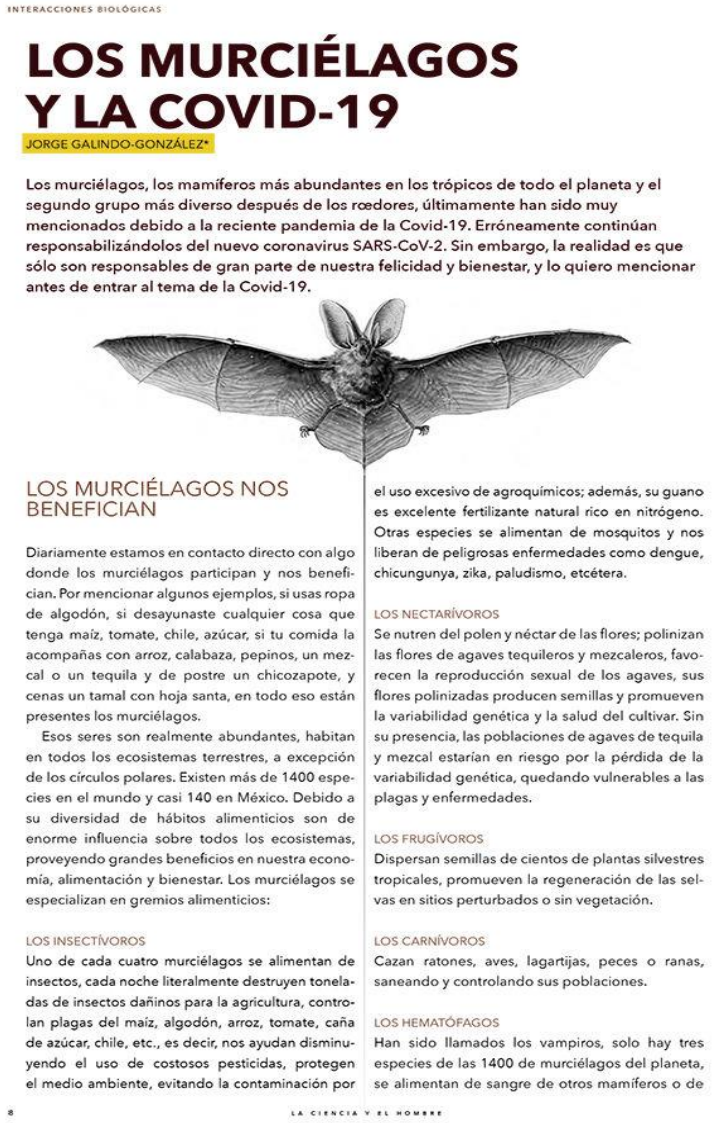
They are presented to a limited audience and for a short time.

Scientific talks and meetings.

Scientific communication channels

Formal channels	Informal channels
Potentially large audience.	Restricted public with limited access.
Information can be permanently saved and retrieved.	Information is not generally saved or retrieved.
The information is relatively old.	The information is recent.
Selection of the channel and content is the initiative of the user.	Selection of the channel and content is the initiative of the user.
Little feedback for the author.	A lot of feedback for the author.

Scientific dissemination articles



INTERACCIONES BIOLÓGICAS

LOS MURCIÉLAGOS Y LA COVID-19

JORGE GALINDO-GONZÁLEZ*

Los murciélagos, los mamíferos más abundantes en los trópicos de todo el planeta y el segundo grupo más diverso después de los roedores, últimamente han sido muy mencionados debido a la reciente pandemia de la Covid-19. Erróneamente continúan responsabilizándolos del nuevo coronavirus SARS-CoV-2. Sin embargo, la realidad es que sólo son responsables de gran parte de nuestra felicidad y bienestar, y lo quiero mencionar antes de entrar al tema de la Covid-19.

LOS MURCIÉLAGOS NOS BENEFICIAN

Diariamente estamos en contacto directo con algo donde los murciélagos participan y nos benefician. Por mencionar algunos ejemplos, si usas ropa de algodón, si desayunaste cualquier cosa que tenga maíz, tomate, chile, azúcar, si tu comida la acompañas con arroz, calabaza, pepinos, un mezcal o un tequila y de postre un chicozapote, y cenas un tamal con hoja santa, en todo eso están presentes los murciélagos.

Esos seres son realmente abundantes, habitan en todos los ecosistemas terrestres, a excepción de los círculos polares. Existen más de 1400 especies en el mundo y casi 140 en México. Debido a su diversidad de hábitos alimenticios son de enorme influencia sobre todos los ecosistemas, proveyendo grandes beneficios en nuestra economía, alimentación y bienestar. Los murciélagos se especializan en gremios alimenticios:

LOS INSECTÍVOROS

Uno de cada cuatro murciélagos se alimentan de insectos, cada noche literalmente destruyen toneladas de insectos dañinos para la agricultura, controlan plagas del maíz, algodón, arroz, tomate, caña de azúcar, chile, etc., es decir, nos ayudan disminuyendo el uso de costosos pesticidas, protegen el medio ambiente, evitando la contaminación por el uso excesivo de agroquímicos; además, su guano es excelente fertilizante natural rico en nitrógeno. Otras especies se alimentan de mosquitos y nos liberan de peligrosas enfermedades como dengue, chicungunya, zika, paludismo, etcétera.

LOS NECTARÍVOROS

Se nutren del polen y néctar de las flores; polinizan las flores de agaves tequileros y mezcaleros, favorecen la reproducción sexual de los agaves, sus flores polinizadas producen semillas y promueven la variabilidad genética y la salud del cultivar. Sin su presencia, las poblaciones de agaves de tequila y mezcal estarían en riesgo por la pérdida de la variabilidad genética, quedando vulnerables a las plagas y enfermedades.

LOS FRUGÍVOROS

Dispersan semillas de cientos de plantas silvestres tropicales, promueven la regeneración de las selvas en sitios perturbados o sin vegetación.

LOS CARNÍVOROS

Cazan ratones, aves, lagartijas, peces o ranas, saneando y controlando sus poblaciones.

LOS HEMATÓFAGOS

Han sido llamados los vampiros, solo hay tres especies de las 1400 de murciélagos del planeta, se alimentan de sangre de otros mamíferos o de

8 LA CIENCIA Y EL HOMBRE

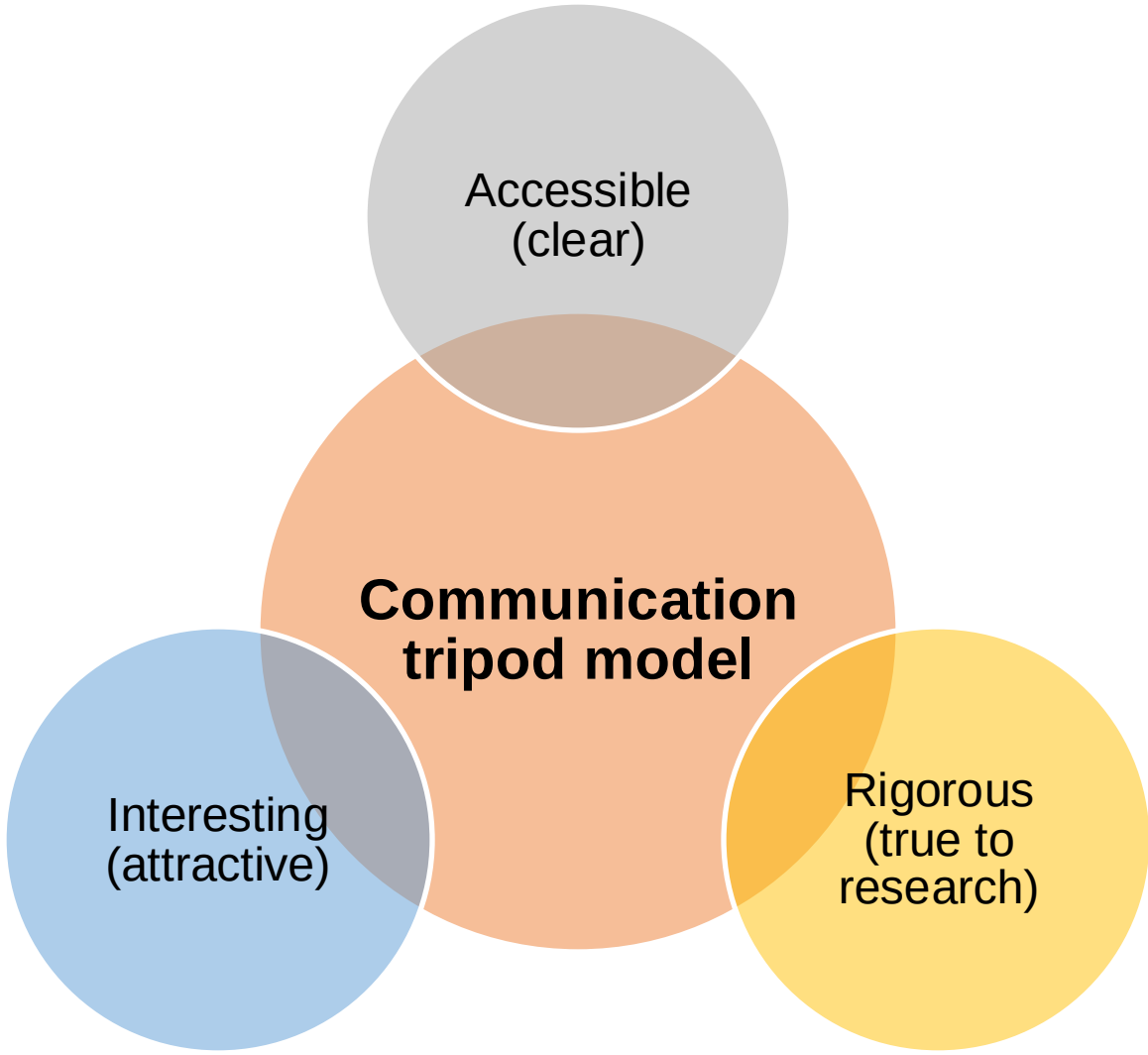
They are informative texts that are based on research projects, scientific articles (papers) or arguments that arise from science.

Its main purpose is to disseminate the research in an accessible way so that it is understandable by the general public.

Scientific dissemination articles

Topic and style

- **The topic** to be discussed must be relevant, attractive and the product of scientific research.
- **The style** must have the structure of an explanatory sequence (spread the information, explain it, interpret it and clarify the ideas).



Scientific dissemination articles

Language

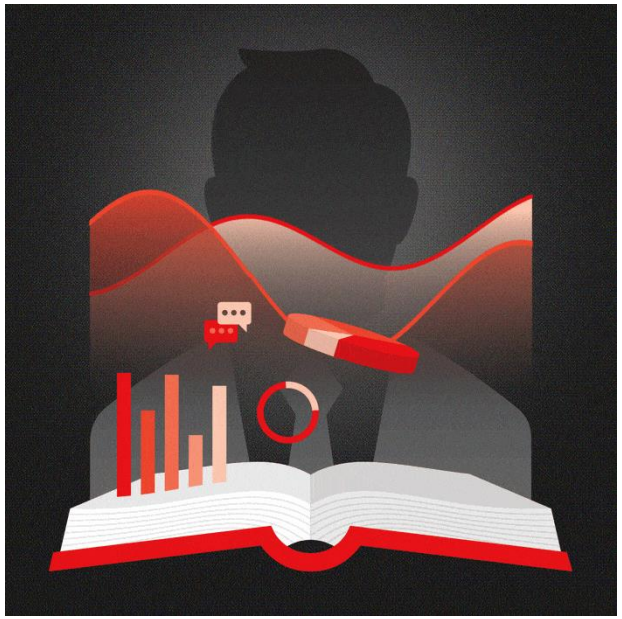
It must be agile, clear and easily understood by the non-specialized reader, avoiding the technical language of your specialty.



Storytelling

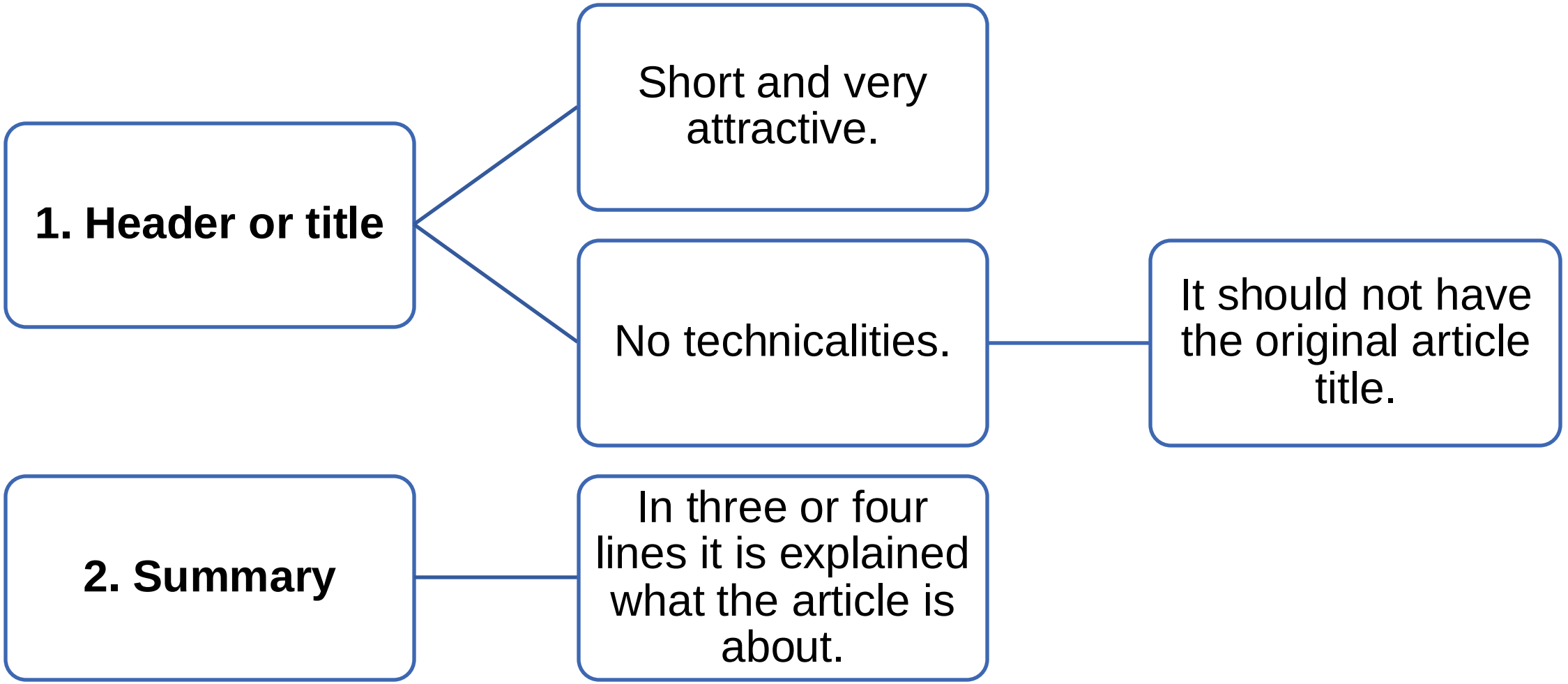
- Telling stories helps to understand the world. The narrative in science is the same.
- The dissemination article tells a story around a research topic.
- It is not about changing the facts, but about changing the way they are presented.

Parts of a dissemination article

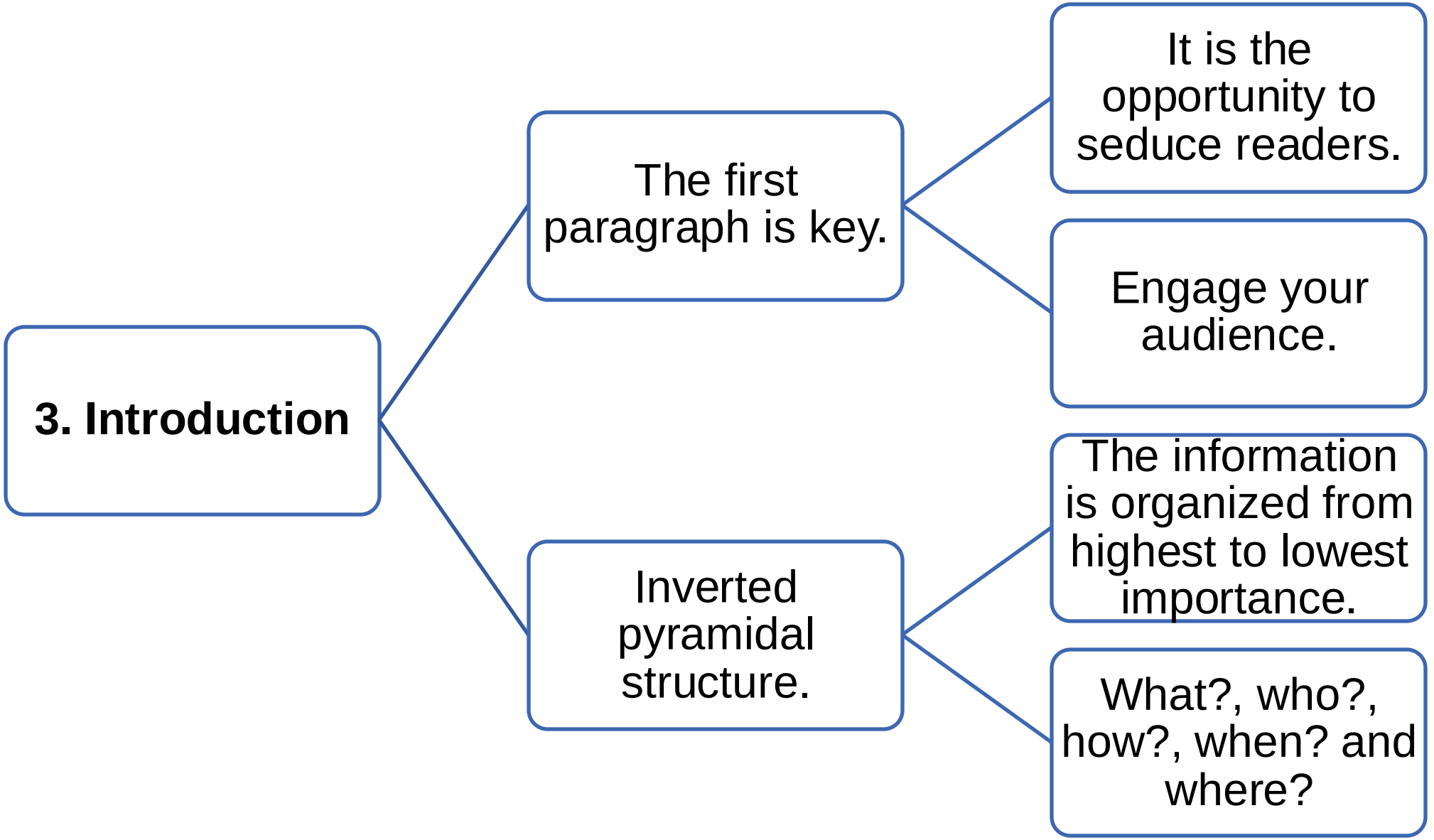


1. Header or title.
2. Summary.
3. Introduction.
4. Development.
5. Conclusions
6. Author's profile

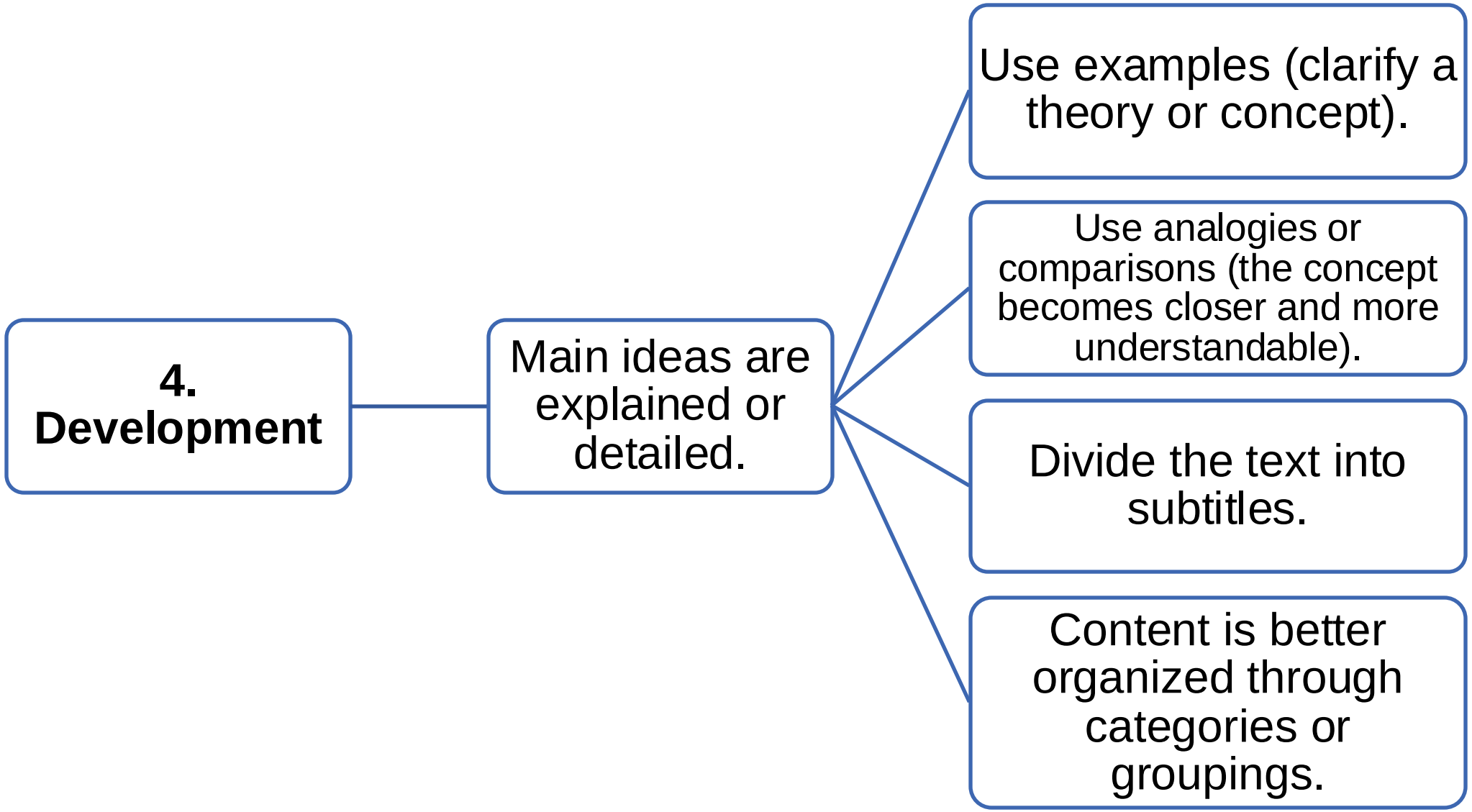
Parts of a dissemination article



Parts of a dissemination article



Parts of a dissemination article



Parts of a dissemination article

5. Conclusions

Make an evaluation on the subject.

- It expresses the possible benefits of the research.
- The closing of the writing is developed.

6. Author's profile

Identification data

- Full name.
- Email.
- Short professional review.

Scientific dissemination articles

Recommendations	Write short paragraphs.
	Use the simple structure of the statement (subject + verb + complement).
	Use synonyms, metaphors, analogies and examples to better explain a complex idea.
	Avoid excessive technicalities. If you need to use a technical term, include a simple definition.
	Humanize the subject, so that the reader can relate the advances of science with their daily life.

Scientific dissemination articles

Support resources

- The following elements may be included in the text, in order to illustrate the article of disclosure:
  - Boxes with complementary information.
  - Photographs.
    Infographics.
    Stadistic charts.
    Illustrations.
    Drawings.
    Glossaries

Scientific dissemination articles

Information photo

Title

Introduction

Capital letter

Development

Photography

Conclusions

Bibliography



AÑO INTERNACIONAL DE LA ASTRONOMÍA

DÓNDE NACEN TODAS LAS ESTRELLAS

La Gran Nebulosa de Orión es tan brillante que se puede distinguir a simple vista, incluso desde lugares con gran contaminación lumínica

La nube o nebulosa se fragmenta en multitud de pequeños cascarones y cada uno formará una estrella

DIARIO CÓRDOBA | ZOCO

Imagen de la Gran Nebulosa de Orión, tomada en un momento en que su tonalidad era roja por la mayor presencia de azufre

Storytelling

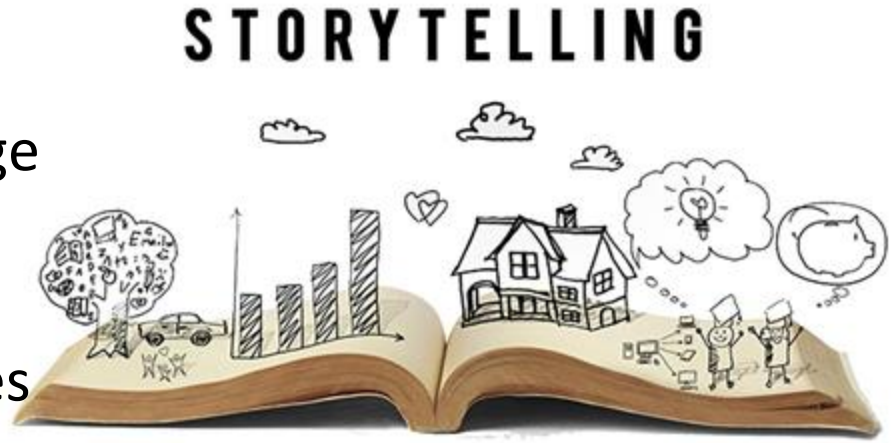
What is it?

- It is a technique that is broken down into two words: history (story) and telling (telling), through which messages are transmitted.

- It is described as a narrative of events with a final message that leaves a learning or concept.

- Experts on the subject say that knowing how to tell stories is an art.

- The goal is to achieve a larger audience, that is, more views and reproductions, but also to have an effect on the people who listen, see or read.



Storytelling

¿What is the objective?

Connect emotionally with the other through a story.

01. Captar la atención

02. Despertar emociones

03. Ganar voluntades

04. Cerrar acuerdos

05. Comprometer acciones

06. Influir y crecer



Storytelling: main elements



1. Message
2. Setting
3. Character
4. Conflict

Storytelling

1. Message

1. Story: a message is transmitted.

2. Telling: the way that message is presented to the audience.

If the message is strong, it may take effect even with a weak saying. But, if the message is poor, you will hardly be able to save your content despite using good techniques.

Texts, stories and conferences that leave the audience enthusiastic for a moment exist in piles, but contents that really impact and make you continue to remember them are few.

2. Setting

Events need a **physical space** in which to place them to continue describing the scenes.

It manages to place the protagonists and scenarios clearly. Imagination and mental translation is easier for those who hear or see the narrative in question.

Events need to happen somewhere. When the environment is well described, it enables the audience to get hooked. In such a way that, it will be from beginning to end.

Storytelling

3. Character

To put together a storytelling, the protagonist or group of people who develop the story that is told must be identified.

The character is the one who runs through the entire story and undergoes a transformation that leads to the transmission of the message.

To reach that final concept, you must overcome your conflict.

The character can be a hero or an antihero.

4. Conflict

It is the **main factor** that causes interest in the audience.

A **very simple conflict does not arouse interest**, since it does not generate identification with the public. Very easy achievements are not usually valued.

It must be elaborate and cannot have an easy solution. In this case, we would have a **romanticized story**. Although it can arouse emotions, it hardly causes identification.

Storytelling: keys to its construction

Authenticity

Lean on past events in our life to be consistent.	Create specific, well-detailed images and strong moments.

Empathy

This feeling will be the element that connects with the public.	Narrate specific examples, look for these elements in the target audience.

Don't pretend

Be authentic at all times.	Remember your body also communicates.

Storytelling: keys to its construction

Emphasis

Generate rhythms in speech, emphasize words, images and energies.	It is recommended not to be arrogant, on the contrary, find common ground with the audience.

Personalize

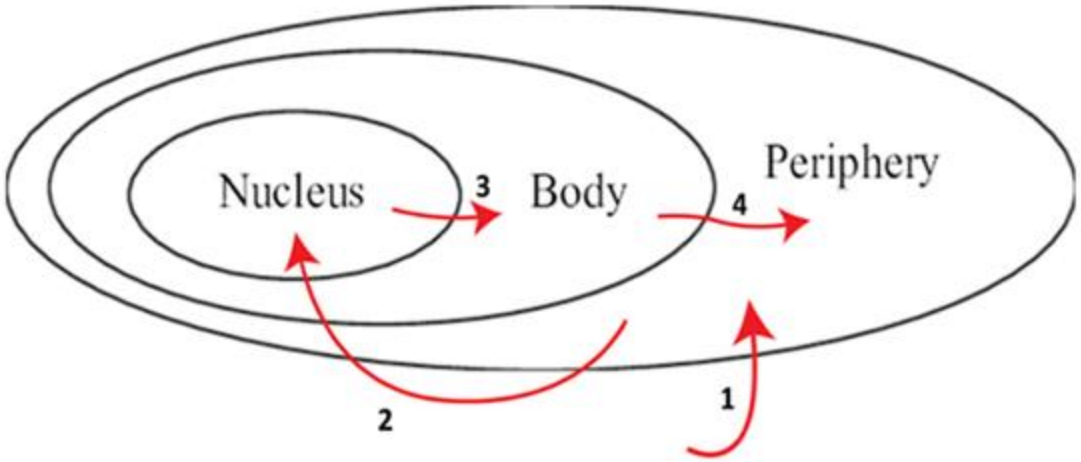
It is recommended not to be arrogant, on the contrary to find common ground with the audience.	Be humble and at the same time speak in a personal way. You must take care of the forms.

Simplify

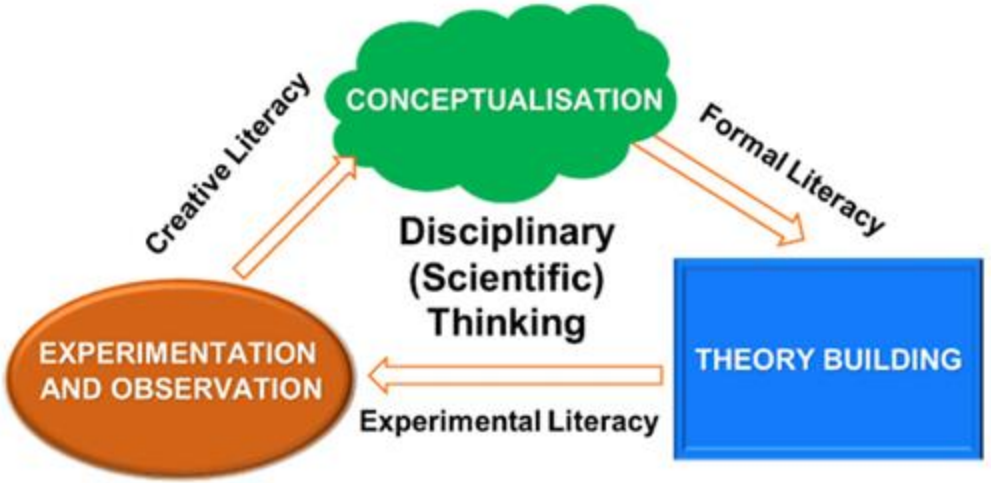
The simpler and shorter the better. Do not embellish the stories with unnecessary facts.	The stories have to have a meaning, that the beginning, middle and end are understood.

Storytelling for Culturo-Scientific Thinking

It focuses on the early development of scientific skills and modern mindset through outreach activities.



(a)



(b)

Storytelling

Conferences



Conferencias de diversos temas, dónde se pone en práctica la importancia de crear un **storytelling.**

Audiovisual production

Autoescuela Sevial-Sevilla (23, julio, 2015) La canción más cara del mundo DGT [Archivo de video]. https://www.youtube.com/watch?v=RatwBxP1V-Q

Ying Liu, R. (12, enero, 2020) Publicidad sobre el medio ambiente [Archivo de video]. https://www.youtube.com/watch?v=QcMn7JpfRVo

Written production



Referencias bibliográficas

01 Deroy Domínguez, D. (2022). *Las revistas científicas y su rol en la difusión del conocimiento científico Scientific Journals and their role in the dissemination of scientific knowledge*. https://orcid.org/0000-0002-5156-7950

02 Zhang, L., Malviya, S., Tandoc, E. C., & Ho, S. S. (2022). Exploring channels and gaps in information dissemination and acquisition among energy scientists and the public: The southeast Asian context. PLoS ONE, 17. https://doi.org/10.1371/journal.pone.0273626

03 Tirado-Espín, A., Cuesta, U., Martínez-Martínez, L., & Almeida-Galárraga, D. (2020). Agenda-setting e inmigración: análisis crítico del discurso y frecuencia en los medios: Estudio descriptivo de investigaciones en revistas científicas desde 2015 a 2020. *Revista Ibérica de Sistemas e Tecnologias de Informação*, (E35), 289-301.

04 Calua, M., Delagado, Y., & López, Ó. (2021). Comunicación Asertiva En el Contexto Educativo : Revisión Sistemática Assertive Communication in the Educational Context : Systematic Review. *Revista Boletín Redipe*, *10*(4), 315–334.

Referencias bibliográficas

05 Ashcraft, L. E., Quinn, D. A., & Brownson, R. C. (2020). Strategies for effective dissemination of research to United States policymakers: a systematic review. In Implementation Science (Vol. 15, Issue 1). BioMed Central Ltd. https://doi.org/10.1186/s13012-020-01046-3

06 Calua, M., Delagado, Y., & López, Ó. (2021). Comunicación Asertiva en el Contexto Educativo : Revisión Sistemática Assertive Communication in the Educational Context : Systematic Review. *Revista Boletín Redipe*, *10*(4), 315–334.

¡Thank you!

ESCANÉAME

Andrés Tirado, Ph.D.
ctirado@yachaytech.edu.ec
098 029 5178

Hacienda San José S/N Urcuquí - Ecuador | Cod. Postal 100119 | Telf.: +593 6 2 999 130 | info@yachaytech.edu.ec
www.yachaytech.edu.ec

